

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester I Examination, 2023-24

Paper: GLB-101 — Elementary Physical and Structural Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 067572

*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory.*

SECTION — A

Question 1. With a neat diagram of a fold, define different elements of it. Describe different types of fold based on the dip of the axial plane, plunge of the hinge line, and fold profile with neat diagrams. [14]

Question 2. Describe the difference between the conformable and unconformable contacts in a sequence of strata. Describe the types of unconformities with neat diagrams. Discuss the importance of unconformities in geology. [14]

Question 3. Discuss in detail the origin and evolution of the hydrosphere and biosphere. [14]

SECTION — B

Question 4. Write short notes on the following: [2 × 7 = 14]

- (a) Overlap, outlier and inlier
- (b) Translational and rotational movement along faults with neat diagrams

Question 5. Discuss with suitable examples: [2 × 7 = 14]

- (a) Paleomagnetism and geomagnetic time scale
- (b) Relative methods of age determination

Question 6. Discuss the following: [2 × 7 = 14]

- (a) Dip-slip fault and normal fault with diagrams
- (b) Different types of chemical weathering processes

Question 7. Briefly discuss the following: [4 × 3½ = 14]

- (a) Objectives of topographic maps
- (b) Lopolith and phacolith

- (c) Body fossil and trace fossil
- (d) Crater and caldera

SECTION — C

Question 8. Describe or fill in the blanks for each of the following in 2 to 3 sentences. Give suitable diagrams wherever necessary: [14]

- (a) Contour map of a plateau [1]
- (b) Types of scale in Structural Geology [1]
- (c) Importance of dip and strike [1]
- (d) Strike joint [1]
- (e) Upright and vertical folds [1]
- (f) Rake of slickenside present on a fault plane [1]
- (g) Azimuth circle on a clinometer compass [1]
- (h) The point of origin of an earthquake is known as _____. [1]
- (i) Among the commonly found radioisotopes, the longest half-life belongs to the _____ decay series and the shortest half-life belongs to the _____ decay series. [2]
- (j) Dinosaurs first appeared in the _____ period and became extinct in the _____ period. [2]
- (k) The protolith for mica-schist is _____ and that for marble is _____. [2]